

Max - Flow/
Min - Cut

Takeaway from Flow discussion

If there are no such $s \sim t$ path (with positive residual capacity on every edge) in the residual graph G_f , then f is a max flow in G .

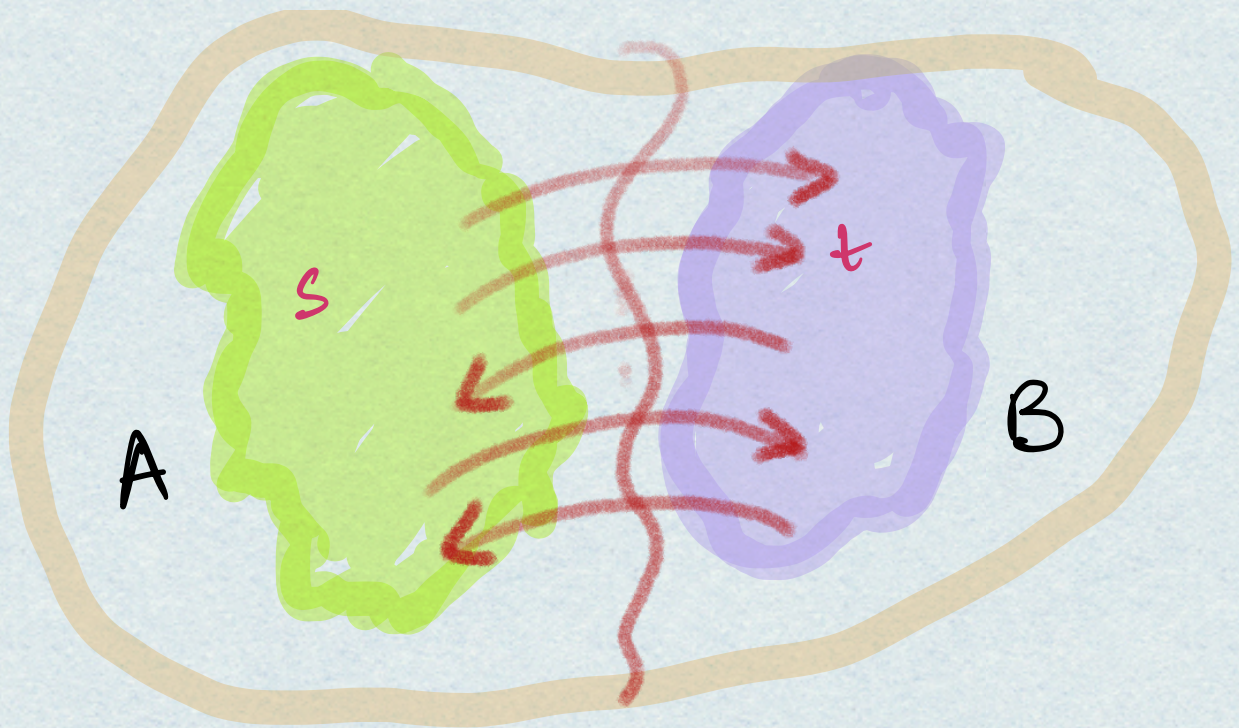
Can we prove a stronger statement out of this? for.

Famous Max-flow /
Min-Cut thm.

(s, t) - Cuts

It's try to create a
sense of "dual" to
flows (will make it
preise)

Defⁿ. (s, t) Cut - An (s, t) -cut
of a graph $G=(V, E)$
is a partition of V into
sets A, B , with $s \in A, t \in B$



Such cuts buckets the
edges of the graph into
four categories :-

- those with both endpoints
in A.
- those with both endpoints
in B

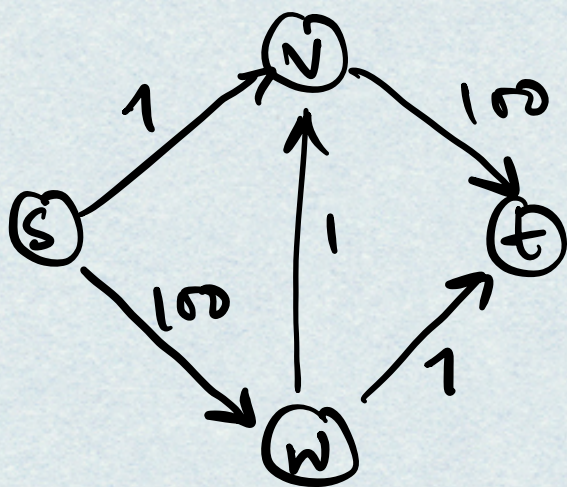
- those sticking out
of A (tail in A
Head in B)

- Those sticking into
A (Head in A, Tail in
B)

The Capacity of an (s,t) cut (A,B) is defined as

$$\sum_{e \in \delta^+(A)} u_e$$

$\delta^+(A)$ - Set of edges sticking in to the source side of (s,t) -cut to do not contribute to its Capacity.



The cut $\{s,w\}, \{v,t\}$ has Capacity 3
(with 3 outgoing edges, with each Capacity 1)

But different cut has diff.
Capacity.

e.g., $\{s\}$, $\{v, w, t\}$
has Capacity 101.

Among all such cuts,
minimum cuts has the
smallest Capacity.

Optimality Conditions for Max Flow

Thm. (Optimality) Let f be a flow in a graph G .

The following are equivalent

1. f is a maximum flow of G .
2. there is a (s, t) -cut S, t the value of f equals the Capacity (A, B) .
3. there is NO s - t path (with +ve residual capacity) in G_f .

Thm. (optimality) asserts
any of the 3 - Statements
implies other 2.

3 $\rightarrow$ 1 (we saw in
last lect.)

Corollary

If f is a flow in G
s.t. the residual network
 G_f has no s-t path.
then f is a Max-Flow.
This $\Rightarrow$ Ford-Fulkerson
Works!

Thm. (optimality) asserts
any of the 3 - statements
implies other 2.

3 $\rightarrow$ 1 (we saw in
last lect.)

Proof :-

Chain of implications

2 $\rightarrow$ 1

1 $\rightarrow$ 3

3 $\rightarrow$ 2

2 $\longrightarrow$ 1

claim, for every flow f
& every (s, t) -cut (A, B)

value of $f \leq \text{Capacity of } (A, B)$



no flow value (●) is $>$
a cut value (●)

why?

Suppose 2 holds.

By the claim ● appears to
the right of ●

Intuitively, feels correct.
But here is an Algebraic proof.

Fix f and (A, B)

By defⁿ:

$$\text{value of } f = \sum_{e \in \delta^+(s)} f_e = \sum_{e \in \delta^+(s)} f_e - \underbrace{\sum_{e \in \delta^-(s)} f_e}_{\text{Vacuous sum.}} \quad (1)$$

Recall the Conservation Constraints

$$\underbrace{\sum_{e \in \delta^+(v)} f_e}_{\text{flow out of } v} - \underbrace{\sum_{e \in \delta^-(v)} f_e}_{\text{flow into } v} = 0$$

for every $v \neq s, t$

Adding (2) corresponds to
all the vertices of $A \setminus \{s\}$
to (1) gives

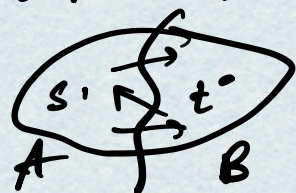
$$\text{Value of } f = \sum_{v \in A} \left(\sum_{e \in \delta^+(v)} f_e - \sum_{e \in \delta^-(v)} f_e \right)$$

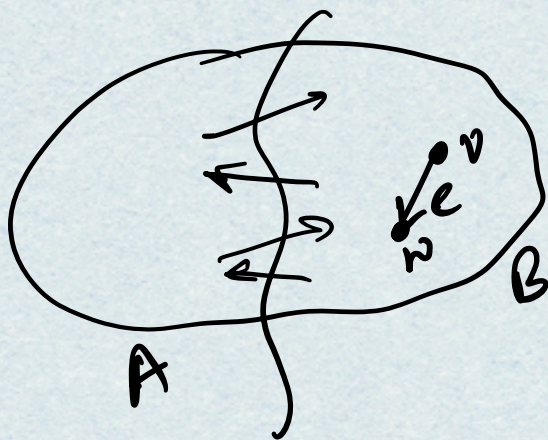
Now let's go back to 3

But from edge-centric P.O.V.
rather than vertex-centric P.O.V.

How much does an
edge contribute to 3?

Well, depends on the bucket it falls. which of
the 4 buckets it falls.





Contributes f_e
once.

(in $\sum_{e \in \delta^+(w)} f_e$)

& $-f_e$ once. (in $-\sum_{e \in \delta^-(w)} f_e$)

Thus, edges inside A

Contribute net zero to 3

Similarly an edge (e) going out
of A Contributes f_e

An edge (e) coming into A
Contributes $-f_e$

Summarizing,

$$\text{value of } f = \sum_{e \in S^+(A)} f_e - \sum_{e \in S^-(A)} f_e$$

⇓
Net flow (flow forward
minus
flow backward)
across every cut is
Exactly Same.

Finally using Capacity Constraints
& the fact that all
flow values are non-
negative

We have

$$\begin{aligned}
 \text{value of } f &= \sum_{e \in \delta^+(A)} \overbrace{f_e}^{\leq u_e} - \sum_{e \in \delta^-(A)} \overbrace{f_e}^{\geq 0} \\
 &\leq \sum_{e \in \delta^+(A)} u_e \quad \text{--- (1)} \\
 &= \text{Capacity of } \underbrace{(A, B)}_{(5)}
 \end{aligned}$$

Completes the proof
of the first implication.

Step 2:

1 $\longrightarrow$ 3

Contr. We prove the Contrapositive

Suppose f is a flow s.t

G_f has an $s \sim t$ path P

the residual Capacity.

As in Ford-Fulkerson Algo.

We augment along P to

produce a new flow f'

with strictly larger value.

this shows f is not
Max flow.

Step 3: $3 \longrightarrow 2$

Final Step is short. & sleek

Let's define Carefully

$$A = \left\{ v \in V : \text{there is an } s \rightsquigarrow t \text{ path in } G_f \right\}$$

- Start with favorite search
(BFS/DFS)
from s until
we get stuck.

(Search part is more for
the proof; not much
required in actual
alg.)

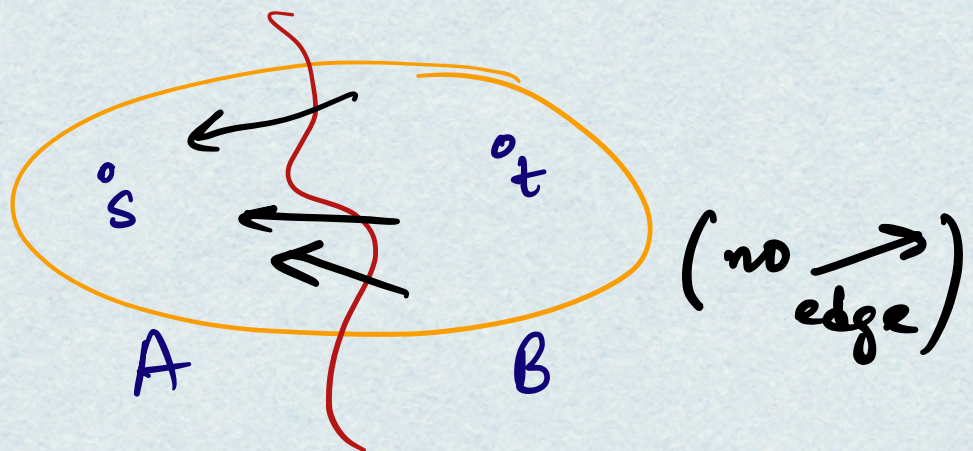
Note,

$(A, V-A)$ is (s,t) -cut

$s \in A$, so s can reach itself
in G_f .

By assumption,

G_f has no $s \sim t$ path,
so $t \notin A$



No edge going out of A
(with +ve residual capacity)

Residual network G_f back
to flow f in the original
network.

1. Every edge striking out of
 A (i.e., in $\delta^+(A)$) is saturated
(means $f_u = u_e$)

For if $f_e < u_e$, for $e \in \delta^+(A)$

Then the residual
network G_f would contain
a forward version of e .

2. Every edge striking into in
 A in G (i.e., in $\delta^-(A)$)
is zapped out ($f_u \geq 0$)

For if $f_e < u_e$ for $e \in \delta^-(A)$

then the residual network
would contain a
forward version of e .

Points 1 & 2 implies

value of $f = \text{Capacity of}$
 $(A, V-A)$



Corollary :-

(Max Flow / Min Cut)

In every network,
max value of a flow =
min Capacity of an
 (S, T) cut.

Proof -

First part of the
proof of the Thm.
implies max. value of
a flow can't exceed the
min Capacity of an (S,t)
-cut.

Third part of the
proof implies that
there can't be a
gap betⁿ max. Flow
& the min cut
Capacity.

Alg. Consequence

Min Cut reduces to
the Max Flow.

Corollary :- Given a max flow,
min cut can be
computed in linear time.

Proof :- Use BFS/DFS to
compute in linear time.

the set A from the third
part of the proof of the thm.

proof shows $(A, V-A)$ is
a min cut. 

Hence, in practice
min cuts can be
typically be computed
using a min. flow
+ the reduction.

